

Reviewer corrections from v0.2 to v0.3

Scope. This note records the corrections made between `paper_B_chordal_toolkit_technical_draft_v02.tex`, `paper_C_fractional_chordal_technical_draft_v02.tex`, and the new v0.3 technical drafts.

The purpose of v0.3 is not to overstate the proof status. It makes the global fractional assembly more robust and explicitly records the remaining local/certificate work in Paper B.

1. Summary matrix

Reviewer / Codex point	Status in v0.2	v0.3 action	Resulting status
Weighted Heavy-Bag bound	Incorrect for arbitrary weights	Replaced unweighted M_* by weighted bag maximum M_x^*	Corrected
Uniform debits for $s_{\alpha} =$ 0,1	Undefined division by $\text{binom}(s_{\alpha},2)$	Added debit-active condition $s_{\alpha} \geq 2$; set active debit to zero otherwise	Corrected
Multi-core / laminar bookkeeping	Present as ownership convention, too compressed	Added explicit retained-core ownership ledger lemma and proof in Paper C	Clarified / strengthened
Terminal classification / promotion completeness L ₄ certificate	Stated as toolkit interface	Kept conditional on Paper B; flagged thresholds as referee-sensitive	Still needs final Paper B package
Core- Fan/Farkas theorem	Claimed as certificate theorem	Added note requiring rational data and exact verifier	Needs re- producible package
Rounding Haxell- Rodl/Yuster	Imported by Paper C	Added debit-active interface and proof-roadmap warning	Needs final normaliza- tion proof
	Not part of Paper C	Left to Paper D	Standard external input

2. Correction A — Weighted Heavy-Bag Lemma

Problem in v0.2

The v0.2 statement said, for arbitrary nonnegative weights,

$$[\{T_u T_v\} x_u x_v M^*_{vx_v}.]$$

This is false when M^* is the unweighted maximum bag size. For example, in K_2 , $M^* = 2$, and weights $x_1=x_2=100$ give left side 10000 and right side 400.

v0.3 correction

Define

$$[M_x^* = t\{v:tT_v\}x_v.]$$

Then the correct bound is

$$[\{uvE(G)\}x_u x_v M_x^*_{vx_v}.]$$

The unweighted statement remains

$$[|E(G)|M^*_n.]$$

Proof idea

Use a perfect elimination ordering. For each vertex, its later neighbours form a clique and hence lie in one bag. Their total weighted mass is at most M_x^* . Orient edges forward in the elimination order and sum.

3. Correction B — Guard uniform debits for small subcores

Problem in v0.2

Paper C defined

$$[z_{\{ij\}} =]$$

for all boundary groups. This is undefined for $s_\alpha = 0$ or $s_\alpha = 1$.

More importantly, when $s_\alpha \leq 1$, the retained subcore contains no core edge ij , so there is no triangle of the form uij that can realize a quadratic boundary debit.

v0.3 correction

Define a boundary group as **debit-active** only when

$$[s = |S|.]$$

Set

[$T_- =$]

Then define

[$z_{\{,ij\}} = , ijS_- ,$]

only for debit-active groups.

Meaning

Groups with one-vertex or zero-vertex subcores are not used for quadratic core-edge triangle savings. Their edges are either handled by internal local packings or left unused and charged to the residual ledger. This weakens the packing but avoids an invalid formula.

4. Correction C — Multi-core / retained-core bookkeeping

Problem in v0.2

The ownership convention was present but too compressed. It could be read as if retained cores had to be disjoint, or as if multi-core overlaps were hidden in lower-order terms.

v0.3 correction

Paper C now contains a standalone lemma:

Retained-core ownership ledger. Run the nearest-heavy recursion with the retained-core convention. Then every edge has a single active owner: internal, boundary, core, or unused. If an edge lies in multiple retained cores, it is owned by the deepest retained core containing both endpoints. Retained-core edges are excluded from descendant relative edge sets.

Key proof point

The clique-tree recursion is laminar. If a vertex support meets two child components after splitting at a bag b , then by connectedness the support contains b . Therefore that vertex is retained as boundary/core, not private in both children. This gives disjoint private sets and prevents duplicate internal ownership.

Edges inside overlapping retained cores are assigned to the deepest retained core containing both endpoints. Descendants cannot reuse those edges because retained-core edges are removed from relative edge sets.

Result

Nested or overlapping retained cores are allowed. What is single-valued is not vertex membership, but **edge-capacity ownership**.

5. What remains conditional after v0.3

v0.3 makes the global fractional assembly more defensible, but it does not by itself attach all local finite proof data.

The remaining referee-sensitive items are:

1. **Promotion completeness / terminal classification.**

Paper B must give exact thresholds separating bounded local residues from long-core or overload promotion events.

2. **L₄ certificate package.**

The public repository should include the rational certificate, exact verifier, type list, edge atoms, triangle atoms, rational weights, and ownership mode.

3. **Core-Fan/Farkas normalization.**

Paper B must spell out the capacity normalization, compressed-fan reduction, and Hall/Farkas dual argument.

4. **Integral rounding.**

Paper D should explicitly cite Haxell–Rodl / Yuster for $[_3(G)_3^{*(G)-o(n^2)}]$ for the fixed family $\{K_3\}$.

6. Suggested response to reviewer

Thank you — this is a fair reading of v0.2. The multi-core bookkeeping was not intended to be hidden inside lower-order terms. The intended mechanism is the retained-core ownership ledger: every edge is assigned to a unique active account — internal, boundary, retained-core, or unused — and retained-core edges are explicitly removed from descendant relative accounts. In v0.3 we promote this convention to a standalone lemma in Paper C.

We also agree with the two Codex comments. The weighted Heavy-Bag statement must use the weighted bag maximum M_x^* , not the unweighted M_* . The uniform boundary-debit formula must be restricted to debit-active groups with $s_alpha \geq 2$; groups with $s_alpha \leq 1$ have no core edge on which a triangle u_{ij} can realize the debit, so they are assigned to the residual ledger.

The remaining local items are promotion completeness, the reproducible L₄ certificate package, and the final Core-Fan/Farkas normalization proof. The global fractional assembly in Paper C is now stated conditionally on those Paper B inputs.

7. Commit suggestions

Fix weighted Heavy-Bag lemma using weighted bag maximum

Restrict uniform boundary-debit realization to debit-active subcores

Add retained-core ownership ledger for recursive multi-core bookkeeping

Record v0.3 proof-status audit and reviewer corrections

8. Additional v0.3 clarification — feasibility vs quantitative telescoping

A later reviewer-sensitive nuance was incorporated into the final v0.3 Paper C file before sending.

Point clarified

The **Retained-core ownership ledger** proves feasibility: every edge has one active owner, so local and boundary fractional triangle assignments patch without double-loading edge capacity.

It does **not**, by itself, prove the quantitative bound

$$[|E(G)| - 2_3^{*(G)} n^2/6 + o(n^2).]$$

That quantitative inequality depends on a separate telescoping argument for retained cores, child relative accounts, and terminal residuals.

v0.3 action

Paper C now contains a separate section:

Quantitative retained-core telescoping.

The new lemma states that if each local Paper B tool supplies the local potential inequality

$$[(R) F(m(R)) - \sum_j F(m(R_j)) + (R), F(x) = x^2/6,]$$

and terminal regions satisfy

$$[(R) F(m(R)) + (R),]$$

then summing over the laminar recursion gives

$$[\{R\}(R) F(n) + \{R\}(R).]$$

Together with the lower-order ledger, this gives

[$+O(n^2)+O_-(n)$.]

Resulting status

The recursive multi-core obligation is now split into two explicit pieces:

1. **Feasibility:** no edge is used by two active packing accounts.
2. **Quantitative accounting:** no quadratic residual/capacity term is counted twice in the potential telescoping sum.

The first is proved by the retained-core ownership lemma. The second is proved by the new quantitative telescoping lemma, conditional on the local Paper B potential inequalities.

This avoids overclaiming what the ownership ledger alone proves.

9. Updated honest status after the final v0.3 adjustment

The global Paper C assembly is now much cleaner: it has a feasibility ledger, a quantitative telescoping ledger, guarded boundary debit realization, and the corrected weighted Heavy-Bag input.

The remaining deep item is still **promotion completeness / terminal classification** in Paper B: exact thresholds and a no-escape proof that every terminal residue either falls into a certified bounded case or triggers promotion. The `L_4` package and Core-Fan/Farkas proof also need to be made reproducible/referee-ready, but the structural bottleneck in Paper C is now explicitly separated and no longer hidden inside “lower-order bookkeeping.”